

ניתוח ועיצוב מערכות תוכנה - עבודת הגשה 2

3.2 - דרישות:

ID	Module	FR/NFR	Desc	Priority	Risk	Status
1	HR	func	The system will store the following details for each employee: employee ID, name, bank account number, employment start date, employment type, salary type, salary amount, vacation days, employment status, branch id	MH	low	imp
2	HR	func	The system must allow only the HR manager to manage employee records, including adding new employees, updating existing employee details, and viewing employee information	MH	low	imp
3	HR	func	The system must keep records of employees who are no longer employed	NTH	low	imp
4	HR	func	The system will allow employees to submit constraints indicating which days and hours they are available to work for the upcoming week.	MH	high	imp
5	HR	func	The system should allow the HR manager to set a deadline for employees to submit their availability.	NTH	low	imp
6	HR	func	The system will assume that employees are available for all shifts unless constraints are submitted.	NTH	low	imp
7	HR	func	The system will allow employees to submit preferences for the upcoming week, indicating: willingness for double shifts and a preferred day off.	NTH	low	imp
8	HR	func	The system should allow double shifts only for employees who	NTH	high	imp

			have requested them in their preferences.			
9	HR	func	The system should track which roles each employee is certified to work. Only the HR manager is authorized to add or change these certifications.	MH	high	imp
10	HR	func	The system will allow the HR manager to assign shifts to employees.	MH	high	imp
11	HR	func	The system will document the history of all shift assignments	NTH	low	imp
12	HR	func	The system will ensure a manager will be selected for every shift	MH	low	imp
13	HR	func	The system must allow the HR manager to define the required roles and the quantity of employees needed for each role per shift.	MH	high	imp
14	HR	func	The system must ensure that employees are only assigned to roles they are certified.	MH	low	imp
15	HR	func	The system will support 2 types of shifts: morning and evening shift	MH	low	imp
16	HR	func	The system should ensure employees are only assigned to shifts that match their constraints. If no other employees are available for a role, the manager may manually override an employee's constraints using a special approval, which requires the employee's consent.	MH	high	imp
17	HR	func	The system will allow only shift managers to take on another role in a shift.	MH	high	imp
18	HR	func	The system will ensure that for every shift in which a delivery is scheduled to arrive, an employee certified as a warehouse worker is assigned to that shift to receive	MH	Low	imp

			it in every branch that is a source/destination in that delivery			
19	HR	func	The system will allow the HR manager to see all deliveries which currently have a warehouse or driver shortage	NTH	Low	imp
20	HR	func	The system must enforce that each employee is assigned to exactly one branch. The HR manager can only assign an employee to shifts that take place within their designated branch.	MH	high	imp
21	Transport	func	The system MUST store all deliveries for tracking and history purposes	MH	Low	To Implement
22	Transport	func	The system MUST allow the Transport manager to assign a driver, registered as an employee, to a delivery, and will ensure that every delivery added to the system has a driver assigned to it.	MH	Low	To Implement
23	Transport	func	The system MUST ensure that the assigned driver has a valid license for the selected truck	MH	Medium	To Implement
24	Transport	func	The system MUST allow the Transport Manager to assign a truck to a delivery	MH	Low	To Implement
25	Transport	func	The system MUST store truck details including license number, model, net weight and maximum carrying weight	MH	Low	To Implement
26	Transport	func	The system MUST store driver details including name and license type	MH	Low	To Implement
27	Transport	func	The system MUST support multiple destinations per delivery	MH	Medium	To Implement
28	Transport	func	The system MUST store site details (source/destination) including address, contact person and phone number	MH	Low	To Implement
29	Transport	func	The system MUST distinguish	MH	Low	To

			between different site types: suppliers and store branches			Implement
30	Transport	func	The system MUST assign each site to a delivery zone	MH	Medium	To Implement
31	Transport	func	The system MUST record the truck weight before departure and at every destination of a delivery	MH	Low	To Implement
32	Transport	func	The system MUST validate that the total weight does not exceed the truck's maximum capacity at any point	MH	High	To Implement
33	Transport	func	The system MUST generate an alert if the weight exceeds the allowed maximum	MH	Medium	To Implement
34	Transport	func	The system MUST generate a uniquely identified delivery document per destination including transported items	NTH	Medium	Not Implemented
35	Transport	func	The system will allow the Transport Manager to modify a delivery in case of a weight violation, by replanning the trip, removing or replacing a destination, replacing the truck, or removing part of the cargo; any such change must be reflected in the delivery document.	NTH	High	Not Implemented
36	Transport	func	The system will ensure that the driver assigned to a delivery is also scheduled in the employee shift roster for the shift during which the delivery is to take place.	MH	High	imp

Non-Functional Requirements:

ID	Module	FR/NFR	Desc	Priority	Risk	Status
37	both	NFR	The system MUST be implemented in Java	MH	low	
38	both	NFR	The system MUST store the data by using a	MH	high	

			database			
39	both	NFR	The system MUST have data upon starting in each of the table			

Changes from assignment 1:

- Added branch id to details that must be stored for each employee
- Added new requirements (18,19,20,36)
- Added to requirement 21 that drivers must be registered as employees and working in a shift in order to be assigned to a drive
- Added that truck weigh-in will occur not only in at the source but in every destination as well
- Added to requirement 35 the ways the system allows to handle weight violations
- Added HR module to requirements 37 & 38
- Changed requirement 38 from storing data in memory to storing in a database
- deleted old requirement that the system must allow start-up without initial data.

Definitions:

Term	Definition
Employee	A worker within the organization, identified by a unique ID, possessing specific employment terms (salary, vacation days, etc.), and assigned strictly to one parent Store Branch.
HR manager	An authorized system user responsible for managing employee records, tracking certifications, setting shift requirements, and generating the weekly shift roster.
Transport Manager	An authorized system user responsible for scheduling deliveries, managing trucks and sites, assigning drivers, and resolving weight violations.

Shift	A designated block of working hours (Morning or Evening) at a specific branch where a defined quantity of employees certified in specific roles is required.
Shift Manager	A mandatory role required for every active shift. This is the only role authorized to take on an additional secondary role within the same shift.
Constraint	An availability restriction submitted by an employee indicating specific shifts they cannot work for the upcoming week. Employees cannot be scheduled for shifts they have constraints for.
Preference	An availability wish submitted by an employee indicating specific shifts they don't want to work in the upcoming week. Employees can be scheduled for shifts they have submitted they don't prefer.
Certification	A qualification assigned to an employee by the HR Manager that authorizes them to fulfill specific job roles (e.g., Cashier, Shift Manager, Warehouse Worker, Driver).
Delivery	A planned transport log consisting of a single source point, one or more destinations, a designated truck, and an assigned driver.
Site	A physical location within the transport network assigned to a specific delivery zone. Sites are classified as either Suppliers or Store Branches .
Supplier	A type of site that acts exclusively as a source or destination in a delivery
Store Branch	A type of retail site that acts as a destination (or source) for deliveries, has its own employee shifts, and requires on-site certified staff to handle cargo.
Driver	An employee certified with a valid driving license corresponding to the assigned truck's classification, who must be active in the shift roster during the delivery.
Warehouse Worker	An employee certified to handle cargo. Their assignment to a shift is mandatory whenever a delivery is scheduled to arrive or depart at that branch.

Delivery Document	A uniquely identified digital document generated per destination that lists the transported items and quantities.
Weight Violation	A critical system state triggered when the gross weight of a truck (empty truck weight + cargo) exceeds its maximum legal carrying capacity.

Open Questions:

#	Topic	Issue	Effect on Current Implementation	Client Answer/ assumption
1	Employee Availability	What happens if no certified employee is available for a specific role in a shift?	high	The system should keep searching for a match. If an employee's constraints cannot be met, the manager can still assign them to the shift using a special approval. We can assume there are enough employees to cover all roles.
2	Multi-Role Assignment	Can an employee perform two different roles in the same shift?	medium	Only the shift manager can take on another role in a shift.
3	Minimum Staffing Requirements	Is there a minimum number of employees required per shift?	low	Only a shift manager is required.
4	Employee's details	Is it possible to add a new employee, delete an employee, or update existing details?	high	Yes, but only the HR manager is authorised to do so.
5	Availability Submission Method	Who is responsible for submitting employee availability into the system? What is the availability's format - list of available days and hours or per shift?	high	The employee submits constraints for next week's shift scheduling. The default setting is that the employee is available to work any shift. list of available days and hours
6	Shift Change	Is it possible to change an existing assignment?	high	Yes, but only the HR manager is authorised to do so.

7	Shift Swapping permission	Is it possible for employees to exchange shifts with one another or is only the HR manager allowed?	medium	Yes, but only the HR manager is authorised to do so.
8	Shift Duration	What is the time range for each shift Morning/Evening)?	low	Yes. The morning shift is between 6:00 and 14:00, and the evening shift is between 14:00 and 22:00.
9	The Role Catalog	What are all the possible job roles in a shift?	low	The possible job roles are: cashier, shift manager, warehouse. There can be more in the future.
10	System Capacity	Is there a maximum number of employees that the system needs to support?	medium	No, there is no limit.
11	Data Retention Policy	How far back should the shift history be preserved?	high	All shift history must be preserved.
12	Shift Staffing Requirements	Does the HR Manager define the required roles for each shift individually, or is there a fixed template for Morning and Evening shifts?	high	The HR manager must manually define different requirements for specific shifts.
13	Employment Terms	What are the employment terms details that must be saved for every employee in the system?	medium	The details of the terms of employment are: start date, scope of position (part-time, full-time), global salary, hourly wage, and vacation days
14	Truck weight	Does the recorded truck weight before departure represent the total weight (truck + cargo) or only the truck's net weight?	High	Assume it represents the total actual weight and validate against maximum capacity
15	Driver licenses	What defines compatibility between a driver's license and a truck? Is it based on truck type, model, or another classification?	High	Trucks have a required license type
16	Items in	Should the transport	High	Assume only basic item data is

	transport	module manage full item details, or only basic data (e.g., item ID, name, quantity)?		stored
17	Delivery zones	How are delivery zones defined, and can a site belong to more than one zone?	Medium	Assume each site belongs to exactly one delivery zone
18	Route planning	Does the system need to optimize the delivery route between destinations, or only store the user-defined order?	High	Assume no route optimization is required at this stage
19	Weight violation handling	When overweight is detected, should the system automatically suggest a solution, or only alert the user?	High	Assume the system only alerts the user
20	Delivery updates	Can a delivery be modified after creation (driver, truck, destinations), or only before departure?	Medium	Assume editing is allowed only before departure
21	Delivery document	What exact fields must appear in the delivery document for each destination?	Low	The document ID, the destination and the transported item list
22	Site roles	Can a site act as both source and destination, and are there restrictions based on site type (supplier vs branch)?	Medium	Assume both roles are allowed but sites are distinguished by type
23	Document numbering	Is the delivery document identifier required to be globally unique across the system or only within a specific delivery?	Low	Assume globally unique identifiers

